A1

PICO 2 SC1632 SOCKETED

DOCK_TX	1	26	SPI_MISO
DOCK_RX	2	27	MEDIA_ARM
MCU_MOTOR_A_N	4	29	INTERLOCK_OK
MCU_DS_B_N	5	31	MEDIA_ERROR_N
MCU_DS_A_N	6	32	AUDIO_PWM
MCU_MOTOR_B_N	7	34	DOCK_BUSY
MCU_DIR_N	9	3	GND
MCU_STEP_N	10	4	GND
MCU_WDATA_N	11	13	GND
MCU_WGATE_N	12	14	GND
MCU_HEAD_N	14	23	GND
MCU_DENSEL	15	28	GND
ASSERT_INDEX	16	33	GND
ASSERT_TRK0	17	20	GND
ASSERT_WP	18	35	3V3
ASSERT_RDATA	20	36	3V3
ASSERT_CHANGE	21	39	VSYS
SPI_CS_N	22	40	VSYS
SPI_SCK	24		
SPI_MOSI	25		

J1

34-PIN PC FLOPPY

GND	1	2	FDD_DENSEL
GND	3	8	FDD_INDEX_N
GND	5	11	FDD_MOTOR_A_N
GND	7	12	FDD_DS_B_N
GND	9	13	FDD_DS_A_N
GND	11	14	FDD_MOTOR_B_N
GND	13	15	FDD_DIR_N
GND	15	16	FDD_STEP_N
GND	17	22	FDD_WDATA_N
GND	19	24	FDD_WGATE_N
GND	21	26	FDD_TRK0_N
GND	23	28	FDD_WP_N
GND	25	30	FDD_RDATA_N
GND	27	32	FDD_HEAD_N
GND	29	34	FDD_CHANGE_N
GND	31		
GND	33		

U1

SN74LVC245AN / 5V-TOL INPUT

5V3	1	5	FDD_MOTOR_B_N
GND	10	15	MCU_MOTOR_B_N
5V3	20	6	FDD_DIR_N
GND	10	15	MCU_DIR_N
FDD_MOTOR_A_N	2	7	FDD_STEP_N
MCU_MOTOR_A_N	19	13	MCU_STEP_N
FDD_DS_B_N	3	8	FDD_WDATA_N
MCU_DS_B_N	17	12	MCU_WDATA_N
FDD_DS_A_N	4	9	FDD_WGATE_N
MCU_DS_A_N	16	11	MCU_WGATE_N

U2

SN74LVC245AN / 5V-TOL INPUT

5V3	1	5	GND
GND	10	NC	
5V3	20	6	GND
GND	NC		
FDD_HEAD_N	2	NC	
MCU_HEAD_N	19	GND	
FDD_DENSEL	3	17	GND
MCU_DENSEL	18	NC	
DOCK_RX_RAW	4	7	GND
DOCK_RX	16	NC	

J2

FLOPPY POWER 5V GND GND 12V

VS_IN	1	GND	3
GND	2	NC	

Q4

TN0702 / FDD INDEX

GND	1	FDD_INDEX_N	3
ASSERT_INDEX	2		

Q5

TN0702 / FDD TRK0

GND	1	FDD_TRK0_N	3
ASSERT_TRK0	2		

Q6

TN0702 / FDD WP

GND	1	FDD_WP_N	3
ASSERT_WP	2		

Q7

TN0702 / FDD RDATA

GND	1	FDD_RDATA_N	3
ASSERT_RDATA	2		

Q8

TN0702 / FDD CHANGE

GND	1	FDD_CHANGE_N	3
ASSERT_CHANGE	2		

RN1

8x10k BUSSED / COMMON PINS

VS	1	8	FDD_DIR_N
FDD_MOTOR_A_N	2	7	FDD_STEP_N
FDD_DS_B_N	3	6	FDD_WDATA_N
FDD_DS_A_N	4	5	FDD_WGATE_N
FDD_MOTOR_B_N	5		

RN2

8x10k BUSSED / COMMON PINS

VS	1	NC	8
FDD_HEAD_N	2	NC	7
FDD_DENSEL	3	NC	6
NC	4	NC	5
NC			

RN3

8x1k BUSSED / COMMON GND

GND	1	ASSERT_CHANGE	8
ASSERT_INDEX	2	DOCK_RX_RAW	7
ASSERT_TRK0	3	MEDIA_ARM	6
ASSERT_WP	4	INTERLOCK_OK	5
ASSERT_RDATA	5		

F1

2A FAST / LOW DROP

VS_IN	1	VS	2

D1

1N5819 / USB BACKFEED BLOCK

VSYS	1	VS	2

C1

47uF 16V

VS	1	GND	2

C2

100nF 50V

3V3	1	GND	2

C3

100nF 50V

3V3	1	GND	2

C5

100nF 50V

3V3	1	GND	2

J3

1725708 / CONTACT WIRES 1 TO 7

GND	1	MEDIA_MOSI	5
MEDIA_3V3	2	MEDIA_MISO	6
MEDIA_CS_N	3	NC	7
MEDIA_SCK	4		

J4

DOCK LINK 3V3 UART

GND	1	NC	5
VS	2	3V3	6
DOCK_TX	3	GND	7
DOCK_RX_RAW	4	NC	8
DOCK_BUSY		NC	9

R1

10k SPI CS DEFAULT

3V3	1	SPI_CS_N	2

R2

10k MEDIA ERROR PULLUP

3V3	1	MEDIA_ERROR_N	2

J5

3V3 OUT / INTERLOCK RETURN

3V3	1	INTERLOCK_OK	2

U4

SN74LVC245AN / SWITCHED MEDIA ISOLATION

MEDIA_3V3	1	NC	11
SPI_CS_N	2	NC	12
SPI_SCK	3	NC	13
SPI_MOSI	4	NC	14
MEDIA_MISO	5	SPI_MISO	15
GND	6	MEDIA_MISO	16
GND	7	MEDIA_MOSI	17
GND	8	MEDIA_SCK	18
GND	9	MEDIA_CS_N	19
GND	10	MEDIA_OE_N	20
		MEDIA_3V3	

U6

LP2951ACN / SWITCHED 3V1 100mA

MEDIA_3V3	1	MEDIA_ERROR_N	8
NC	2	NC	7
MEDIA_OFF	3	MEDIA_FB	6
GND	4	VS_MEDIA	5

R4

10k 1%

MEDIA_3V3	1	MEDIA_FB	2

R5

10k 1%

MEDIA_FB	1	GND	2

R6

6K8 FAILSAFE BLEED

MEDIA_3V3	1	GND	2

R7

680R DEFAULT OFF

VS	1	MEDIA_OFF	2

R8

2R2 DISCHARGE

MEDIA_3V3	1	DISCHARGE_C	2

R9

10k MSO DEFAULT

SPI_MISO	1	GND	2

R10

1 OHM CAP ESR

MEDIA_CAP_RETURN	1	GND	2

R11

2k2 ARM GATE

MEDIA_ARM	1	ARM_BASE	2

R12

2k2 INTERLOCK GATE

INTERLOCK_OK	1	LOCK_BASE	2

Q1

IRLZ44N / HARD DISCHARGE

MEDIA_OFF	4	GND	3
DISCHARGE_C	2		

Q2

TN0702 / ARM PERMISSION

PERMISSION_MID	1	MEDIA_OFF	3
ARM_BASE	2		

Q3

TN0702 / INTERLOCK PERMISSION

GND	1	PERMISSION_MID	3
LOCK_BASE	2		

C6

22uF 16V / OUTPUT

MEDIA_3V3	1	MEDIA_CAP_RETURN	2

C7

100nF 50V

MEDIA_3V3	1	GND	2

C9

1uF 50V X7R

VS	1	GND	2

Q9

SUP53P06 / MEDIA INPUT CUT

MEDIA_OFF	1	VS	3
VS_MEDIA	2		

Q10

TN0702 / OE ARM

OE_MID	1	MEDIA_OE_N	3
ARM_BASE	2		

Q11

TN0702 / OE INTERLOCK

GND	1	OE_MID	3
LOCK_BASE	2		

R13

10k OE DEFAULT OFF

MEDIA_3V3	1	MEDIA_OE_N	2

U7

LM386N-1 / SAMPLE AUDIO

NC	1	AMP_OUT	5
GND	2	VS	6
AUDIO_IN	3	AMP_BYPASS	7
GND	4	NC	8

J6

SPEAKER B OHM / + GND

SPK_PLUS	1	GND	2

R14

47k AUDIO ATTENUATOR

AUDIO_PWM	1	AUDIO_LP1	2

R15

2k2 AUDIO BIAS

AUDIO_LP1	1	GND	2

R16

2k2 AUDIO FILTER

AUDIO_LP1	1	AUDIO_LP2	2

R17

10R AUDIO STABILITY

AUDIO_ZOBEL	1	GND	2

C10

22nF AUDIO LP1

AUDIO_LP1	1	GND	2

C11

22nF AUDIO LP2

AUDIO_LP2	1	GND	2

C12

100nF AUDIO COUPLING

AUDIO_LP2	1	AUDIO_IN	2

C15

100nF AMP DECOUPLING

VS	1	GND	2

C16

100nF AUDIO STABILITY

AMP_OUT	1	AUDIO_ZOBEL	2

C13

100uF 16V / SPEAKER DC BLOCK

AMP_OUT	1	SPK_PLUS	2

C14

47uF 16V / AMP RESERVOIR

VS	1	GND	2

C17

22uF 16V / AMP BYPASS

AMP_BYPASS	1	GND	2

R1: Standard IBM-PC 34-pin FDD rear interface: cable twist handled by firmware selection.
5V inputs use 5.5V-tolerant LVC245 at 3.3V. Outputs are TN0702 open drain: ASSERT=1 pulls cable LOW.
J3 is the isolated seven-wire contact output. J5 series seat-NO/eject-NC loop MUST open before disk motion.
J6: 8-ohm speaker, PWM audio on GP27. J4 pin6 reserved: dock lighting via UART.
Pico is a factory-assembled SMD module plugged into TH sockets. Carrier assembly is through-hole.
Legacy floppy encoding/track streaming and native dock protocol require new firmware; no compatibility claim.

PROTOTYPE / NOT RELEASED / electrical and mechanical bench tests required

Sheet: /
File: cyber-drive-controller-r1.kicad_sch

Title: RODDY CYBER DRIVE / CORE CONTROLLER

Size: A1	Date:	**Rev: R1.2 drive audio**
KiCad E.D.A. 10.0.6		Id: 1/1